Assignment: ICT and Its Role in the Field of Education

Assignment Title:

"The Evolution of Computing and Its Impact on Education"

Assignment Purpose:

This assignment will help students demonstrate their understanding of the **foundations of ICT** and **analyze how ICT shapes the modern educational landscape**, aligned with the module's intended learning outcomes (ILOs).

Assignment Instructions:

Prepare a **structured report** (approximately **2000–2500 words**) covering the following tasks. Use **headings, subheadings, diagrams, and real-world examples** from the field of education wherever possible.
Include references where needed.

Assignment Tasks:

Task 1: Evolution of Computers

Compiled by J.K.S.Ruthirangi (Dr. Prof.Study, MA in TE, BEd, HigDip in ICT, Dip in DBA) for CINEC BEd (Hons) Top up Degree.

- Describe the **evolution of the computer** from early mechanical devices to modern systems.
- Identify and explain the **five generations** of computers.
- Focus particularly on **how each generation impacted educational practices** (e.g., calculators in classrooms, computer labs, online learning).

Expected Output:
300–400 words + Timeline or simple visual chart (optional bonus)

Task 2: Basic Computer Architecture and Operations

- Define **basic computer architecture**.
- Explain **how computers work**: input, processing, storage, and output operations.
- Create a **simple labeled block diagram**.
- Give an example of how understanding computer architecture helps teachers/students use computers effectively in education.

Expected Output:
300–400 words + Diagram

Task 3: Basic Components of a CPU and Program Execution

- List and explain the **main components of a CPU** (ALU, CU, Registers, Cache).
- Describe **step-by-step how a simple program** (e.g., a math quiz application) is executed inside a CPU.
- Highlight why it is important for educators to have a **basic understanding** of how computers process tasks.

Expected Output:

300–400 words

Task 4: Computing and Education: Mutual Impact

- Analyze **how computing has changed education** (e.g., virtual classrooms, e-learning, assessment tools).
- Explain how **education influences the development of ICT tools** (e.g., the demand for educational apps, LMS systems like Moodle).

Expected Output:

300–400 words

Include examples like smartboards, online exams, etc.

Task 5: IT and Related Disciplines in Education

- Explain the relationship between **Information Technology** and other fields such as **educational psychology, curriculum design, instructional technology**.
- Use examples showing how **knowledge from other disciplines informs ICT practices in education** (e.g., using cognitive load theory when designing e-learning platforms).

Expected Output:

300–400 words

Task 6: Changes in Application Domains Due to IT in Education

- Explain **how and to what extent ICT has transformed different areas of education**, such as:
    - Content delivery (e-learning, MOOCs)
    - Student assessment (online quizzes, automated feedback)
    - Administration (student information systems)
- Discuss **both positive changes and challenges**.

Expected Output:

300–400 words

Submission Requirements:

- Format: Typed, Times New Roman 12 pt font, 1.5 line spacing.
- File format: PDF or Word document.
- Include a **cover page** (title, student name, registration number, date).
- Cite references in **APA format** (minimum 3 sources).

Evaluation Rubric:

Criteria	Marks
Understanding of Concepts	20
Relevance to Education Field	20
Structure and Presentation	15
Use of Examples and Diagrams	15
Language and Grammar	10
References and Citations	10
Creativity and Critical Thinking	10
Total	100

Reminder:

Compiled by J.K.S.Pathirana (Dr. Prof.Study , MA in TE, BEd, Hig Dip in ICT, Dip in DBA) for CINEC BEd (Hon) Top up Degree.

This is not only a theoretical report — you are expected to **connect all your answers to the field of education** with **real-life examples** (e.g., computers in classrooms, mobile learning apps, etc.).